SECOND REVIEW · ACADEMIC PROTOTYPE · NOT AN OPERATIONAL SYSTEM

Component B — Sentiment, emotion, and stance

How one post is read for feeling, emotion, stance, sarcasm, and the reaction bar.

This brief is for the teammate who presents the reading of each post. The models are used as published. We did not train them.

1. What this agent does

Component B reads the text of one post and writes one result row. It also scores the emoji reaction bar under the post. Those two readings stay separate. A crowd of likes does not rewrite the text label.

2. Where it sits

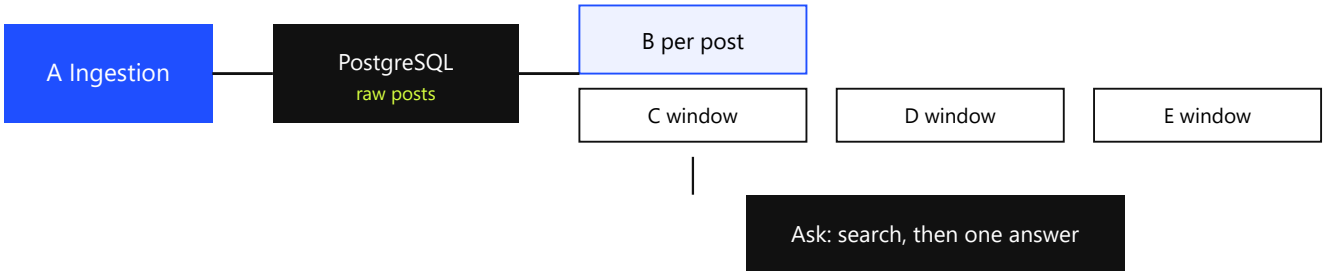


Figure 1. Posts land in Postgres. B reads one post at a time. C, D, and E recompute a time window. Ask searches the short summaries those agents write.

B is the only row runner. A post starts as `pending`. After B, it is `processed`. C, D, and E do not wait for that flag. They read a time window of posts.

3. The four readings

Reading	Method	Output
Sentiment	`cardiffnlp/twitter-xlm-roberta-base-sentiment`	positive, neutral, or negative
Emotion	`SamLowe/roberta-base-go_emotions`, mapped to Ekman emotions plus neutral	joy, anger, sadness, fear, disgust, surprise, neutral
Stance	`valhalla/distilbart-mnli-12-3` against the channel topic	supportive, against, neutral, or unclear
Sarcasm	Rules on text and emoji. No neural model.	A flag only. The sentiment label stays as the model gave it.
Reaction bar	Count of 👍 🔥 and similar marks versus 👎 🤡 and similar marks	A separate score. Cutoff 0.15. Unmapped emoji do not move it.

4. Why the models run one after another

The reference laptop has a 4 GB graphics card. Loading sentiment, emotion, and stance together exceeds that budget. The pipeline loads one model, finishes the batch, releases it, then loads the next. On a machine without a usable GPU the same order runs on the CPU. It is slower. The labels do not change because the device changed.

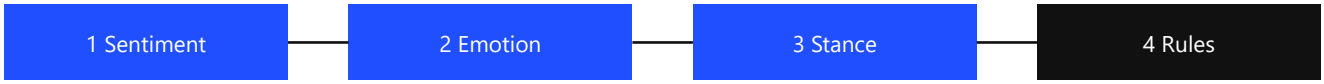


Figure 2. Memory is cleared between the three neural models. Sarcasm and the reaction bar need no graphics memory.

5. Stance rule

DistilBART asks whether the text supports or opposes the channel topic. If the best score is under 0.60, the stance is neutral or unclear. We do not force a side. The larger BART model is only a comparison for the report. It is not loaded in the demo, because it is about 1.6 GB.

6. What to show

Open **Sentiment**. Point at the positive, neutral, and negative bars, then at the reaction chips. Say the reaction row is the audience tapping the post, and the bars are the text model.

> **One sentence.** "Sarcasm lowers our confidence. It never turns a positive label into a negative one."

7. Related parts

Part	Relation to B
Run analysis	The button sends at most 50 pending posts through B, then refreshes C, D, and E. That cap is the viva timing budget, not the size of the archive.
Table	`sentiment_results_b`, one row per post.
API	`GET /api/sentiment/timeline`
Ask	B's short summary can be retrieved later. The raw post is not the search document.

8. Done, and not done

Done	Not claimed
Sequential pipeline, stance threshold, sarcasm flag, reaction score, Sentiment page.	No accuracy percentage. SemEval comparison is Phase 5 and has not been run.
Empty text is skipped. A photo with no words cannot be given a text sentiment.	Not every newly downloaded post has been through B yet. Pending rows stay pending until a run.

> **Do not say** "98 percent accurate", and do not say the reaction bar is the sentiment.

9. Likely questions

Question	Answer
Did you train the model?	No. These are public Hugging Face checkpoints used off the shelf.
Why DistilBART?	About 260 MB. The large MNLI model does not fit the 4 GB card beside the others.
Why is a new channel missing?	A is storage. B has not been run on those pending rows yet.
